JOHNATHAN STOWERS

Macon, GA | (210) 827-5077 | j.stowers1@icloud.com | jstowers1.dev | in/johnathan-stowers | github.com/Jstowers1

EDUCATION

Mercer University Macon, GA
B.S. in Computer Science & M.S. in Software Systems (4+1 Program) Expected May 2028

- **GPA:** 3.82 / 4.00 | **Minors:** Mathematics, Cybersecurity
- **Honors & Awards:** 7th Place — CCSC Competitive Programming Conference (2025), Presidential Scholar, Dean's List
- **Relevant Coursework:** Data Structures & Algorithms, Artificial Intelligence, Graduate Neural Networks (Fall 2026), Linear Algebra, Discrete Mathematics

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, SQL, Bash

Frameworks & Infrastructure: Docker, FastAPI, React, Node.js, GitHub Actions, Git, Linux, PyTorch, NumPy, REST APIs, HTCondor, Tailscale, Nginx

EXPERIENCE

Mercer University Macon, GA
Artificial Intelligence Researcher Sep 2025 – Present

- Engineered high-throughput CNN data pipelines using **Python** and **NumPy** processing large-scale cosmic ray simulation datasets from the IceCube Neutrino Observatory (5,160 optical sensors across 1 km^3 of Antarctic ice).
- Architected a distributed extraction pipeline on an **HTCondor** compute cluster using IceCube's I3Tray framework, transforming gigabytes of raw detector simulations into 10×10 hexagonal-to-square spatial matrices for model training.
- Built automated batch submission scripts for an HTCondor compute cluster, eliminating manual job orchestration across distributed compute nodes.
- Presented research findings and pipeline architecture at Mercer's BEAR Day undergraduate research symposium.

KEY PROJECTS

Homelab Docker Dashboard | *Python, Docker SDK, Tailscale VPN, GitHub Actions* Source Code

- Built a container management dashboard controlling 10+ Docker services across 4 Linux servers via Tailscale mesh network.
- Implemented real-time health monitoring via the Python Docker SDK and CI/CD with GitHub Actions for zero-downtime SSH deployments.

Mechanistic Interpretability Lab | *Python, PyTorch, NumPy, PySide6 (Qt6), pyqtgraph* Source Code

- Built a native Linux desktop application for interactive neural network interpretability research across three modules: superposition simulation, sparse autoencoder (SAE) training, and activation patching.
- Implemented a superposition simulator in pure **NumPy** modeling how N features pack into $M < N$ dimensions, computing interference matrices and reconstruction error via SVD projection.
- Built an SAE training engine in **PyTorch** with L1 sparsity penalties and live loss/sparsity tracking, plus a causal activation patching module measuring layer-wise effects on model outputs.

Custom Hermes Desktop App | *Python, GTK3, WebKit2, SQLite (FTS5), REST APIs* Source Code

- Developed a native Linux desktop AI application wrapping web UI agents inside GTK3/WebKit2 containers with dynamic display orientation detection for multi-monitor workflows.
- Built an embedded knowledge retrieval engine using SQLite Full-Text Search (FTS5), delivering sub-millisecond query responses over a lightweight REST API.

Server Fleet Discord Bot | *Python, Docker, SQLite* Source Code

- Engineered an asynchronous, containerized Discord bot executing 13+ automated administrative and analytics commands across active community channels.
- Implemented fault-tolerant persistent state management using isolated Docker bind-mounts and SQLite database transactions to maintain continuous runtime stability.